



Unit 4 · Outcome 1 · Climate Change

We can measure today. How do we know tomorrow?

KK08

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



We can measure today. How do we know tomorrow?

KK08 Unit 4 · Climate Change · Dot point 8 We can measure today . How do we know tomorrow ? At Cape Wirreanda Climate Observatory, instruments record the planet in real time — a thermometer reads the air, a tide gauge reads the sea, a satellite reads the ice. But no instrument can read the year 2090. To see the future, scientists build models . This dot point is about both halves of that toolkit: direct measurement and modelling . Loading the observatory... Cape Wirreanda Climate Observatory. A weather station, a tide gauge and an orbiting satellite — t

Two tools, two time-frames

02 The framework Two tools, two time-frames Everything in this dot point fits in one little grid. There are two tools for studying climate variability, and we use them across two time-frames . Get this grid straight in your head and the whole topic clicks into place. Direct measurement instruments reading the real world Modelling physics simulated on a computer Past & present The instrumental record — thermometers since ~1850, tide gauges since the 1880s, satellites since the 1970s. This is what has actually happened , recorded as it happened. Run a mode

The four indicators VCAA names

03 What we actually measure The four indicators VCAA names The study design lists four things to track. Memorise them as a set — examiners love a question that says "describe two indicators of climate variability and how each is measured." Each one can be measured directly today and modelled into the future. Global temperature The headline number. Land stations + ocean buoys + satellites, averaged into a single global value reported as an anomaly (departure from a baseline average). DIRECT: station network Rising sea levels Measured by tide gauges (

Global temperature — and the anomaly trick

04 Indicator 1 Global temperature — and the anomaly trick You can't describe Earth's temperature with one thermometer — it's -40°C in Antarctica and $+45^{\circ}\text{C}$ in the Pilbara on the same day. Instead, scientists run a global network : thousands of land weather stations, ships and drifting Argo ocean buoys, and satellites measuring the whole surface. Each reading is turned into an anomaly — how far above or below a long-term average (e.g. 1961–1990) it sits — and the anomalies are averaged into one number. Why an anomaly, not a raw temperature? Anomalies tra

The sea-level lab

05 Indicator 2 · Interactive The sea-level lab Global mean sea level has risen about 20 cm over the last century , and the rate is accelerating. Two physical processes drive it — and one tempting third process does nothing at all . Run the experiment below to see why. Tap each driver and watch the tide gauge on the right. Loading the water tank... Warm the ocean Melt land ice Melt floating sea ice ↵ Tide gauge: 0.0 cm Two real drivers, one impostor. Warming expands the water; melting land ice adds new water. Melting floating ice changes nothing —

Snow & ice — read from orbit

06 Indicator 3 Snow & ice — read from orbit The cryosphere (the frozen part of Earth) is one of the clearest fingerprints of a changing climate, and almost all of it is measured by satellite . We track the extent of Arctic sea ice each September, the mass of the Greenland and Antarctic ice sheets (the GRACE satellites literally weigh them by measuring tiny changes in gravity), the length of retreating glaciers, and the area of seasonal snow cover. What changes How it's measured (direct) What it tells us Arctic sea-ice extent Satellite microwave imaging Declining ~13% per decade in summer — a strong warming signal (and an albedo feedback) Ice-sheet mass GRACE gravity satellites Greenland & Antarctica are losing mass → directly raises sea level Glacier length Repeat satellite & field surveys Near-global retreat → water supply & sea-level impacts Snow cover Satellite optical imaging Shorter snow seasons → affects albedo, water & alpine ecosystems

Local & regional extremes — variability you feel

07 Indicator 4 Local & regional extremes — variability you feel Global averages are smooth, but climate change is mostly experienced as a shift in extremes at a particular place: more frequent heatwaves, more intense downpours, deeper droughts, more dangerous fire-weather days. These are tracked from local station records — counting how often a threshold is crossed and watching that count change over the decades. ★ The key idea — a small shift in the average is a big shift in the extremes Picture the bell curve of daily temperatures shifting just a littl

Modelling — how we read a future we can't measure

08 The other half · Interactive Modelling — how we read a future we can't measure A climate model (a Global Climate Model, or GCM) divides the atmosphere and ocean into a 3-D grid of millions of cells. In every cell, the computer solves the known physics — the flow of energy, air, moisture and heat — then steps forward in time, again and again, building a simulated climate. Press Run below to wrap Cape Wirreanda's world in a model grid and watch the simulation light up. Loading the model grid... Show grid ► Run model Auto-rotate grid ready A Global C

Term	Plain meaning	Exam use
GCM (Global Climate Model)	3-D physics grid of the whole planet	The main modelling tool
Hindcast	Run the model on the past	Validates the model against measurements
Projection	Run the model on the future	Conditional, scenario-based, a range
Emission scenario	An assumed future path of emissions	Low / medium / high → fan of outcomes
Ensemble	Many models/runs together	The spread shows the uncertainty

The Cape Wirreanda dashboard

09 Synthesis The Cape Wirreanda dashboard Put it all together. The observatory measures four indicators directly, today — then feeds those records into models to hindcast the past and project the future.

Temperature Direct: station + buoy + satellite network → global anomaly. Model: projected to rise under all scenarios. Sea level Direct: tide gauges + altimetry. Drivers: thermal expansion + land-ice melt (not sea ice).

Snow & ice Direct: satellites (extent) + GRACE gravity (mass). Declining almost everywhere. Extremes Direct: local station rec

Command-word decoder

◆ COMMAND WORDS

10 Exam technique Command-word decoder The verb in a question tells you exactly what to do. Tap each to expand what an examiner wants — and how it changes your answer for this dot point. Identify / State 1 mk Name it, no explanation. "Identify one direct method used to measure sea-level rise." → "Tide gauges." One word can score the mark. Describe 2–3 mk Say what and how , with detail. "Describe how global temperature is measured." → network of stations/buoys/satellites → anomalies → averaged into one global value. Explain 3–4 mk Give the cause-and-effect

Sort it out — drag each card to the right bin

◇ ACTIVE RECALL

11 Active recall Sort it out — drag each card to the right bin Is each one a direct measurement , a modelling activity, or an indirect / proxy method from the previous dot point? Drag the chips into place, then check. Direct measurement Modelling Indirect / proxy (KK07) s

The P·E·L trainer

🔑 P·E·L WRITING

12 Writing structure The P·E·L trainer For any "explain" or "justify" question, structure each marking point as Point → Evidence → Link . Here it is on a 3-mark question: "Explain how scientists can be confident about future climate projections." P Point State the mechanism directly. Point: Climate models are tested against the past before being trusted with the future. E Evidence Name the specific process or data. Evidence: When a model is run as a hindcast , it reproduces the directly measured temperature record of the 20th century. L Link Tie it back



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Weak vs strong — same question, very different marks

✗ WEAK ANSWER

✗ Weak response — 1/4 As the planet gets hotter, all the ice caps melt and this makes the sea go up . The polar bears lose their ice. Sea ice melting raises the sea level too. This is a big problem for low islands.

✓ STRONG ANSWER & MARKING

Why it loses marks: No mechanism — doesn't distinguish land ice from sea ice, and gets the sea-ice point backwards (the trap!). Misses therm



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

Practice exam — 22 marks

★ PRACTICE & SELF-MARK

14 Exam practice · self-mark Practice exam — 22 marks Have a real go on paper before revealing each answer. Then mark yourself honestly with the buttons — your score builds in the dock, bottom-right.

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

As the planet gets hotter, all the ice caps melt and this makes the sea go up . The polar bears lose their ice. Sea ice melting raises the sea level too. This is a big problem for low islands. Why it loses marks: No mechanism — doesn't distinguish land ice from sea ice, and gets the sea-ice point backwards (the trap!). Misses thermal expansion entirely. Emotive padding ("polar bears") earns nothing.

✓ STRONG / MODEL ANSWER

Land-based ice — glaciers and the Greenland and Antarctic ice sheets — sits on land , so when it melts it adds new water to the ocean, raising sea level. Warming also causes thermal expansion: as seawater heats, it expands and takes up more volume, which is the other major driver. Melting floating sea ice does not raise sea level, because it already displaces its own weight in water (Archimedes' principle). Why it scores 4/4: Mechanism for land ice (1) + thermal expansion named and explained (1) + sea ice identified as the exception (1) + correct reason via displacement (1). Precise, no padding.